WAC
Windows Artefacts Collector

CENTRE D'EXCELLENCE CYBERDEFENSE AEROSPATIALE

Created by the Aerospace Cyber Defence Centre of Excellence (CEC)

Validation report
WAC (Windows Artefacts Collector)

Windows artefact collection software for digital forensic investigation



SYSTEM ARTEFACTS

USER FILES

DIGITAL EVIDENCE

A TOOL IN SUPPORT OF DIGITAL INVESTIGATION

WAC — Windows Artefacts Collector

Validation report

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This document lists the checks that establish the quality of WAC: those WAC runs itself during every collection, the test programs that confront its decoders with a reference, and the automatic checks of its outputs. For each, it states what is compared, against what, the reference result and the defects it has already caught. It also states what these checks do not guarantee. How to run each test is described in the *Testing guide*.

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CHAPTER 1

The approach

1.1 Why valid JSON proves nothing

WAC produces JSON files meant for a forensic report. Their validity says almost nothing about their accuracy: a field read at the wrong offset, a date interpreted in the wrong time zone, a name decoded in the wrong code page, a wrong decompression all give perfectly valid JSON, and wrong. The most serious defects found in WAC were all of that kind. None showed at run time; each only appeared when a value was confronted with an **independent source**.

WAC's validation therefore rests on one principle: every piece of data that can be cross-checked is, by a means that shares nothing with the code under test.

1.2 Four levels of checks

The checks built into WAC (chapter 2): run during every collection, on the examined machine. They refuse inconsistent data rather than publish it, and attest the integrity of the exhibits taken.

The test programs (chapter 3): sixteen executables that confront each decoder with a reference — the Microsoft function it replaces, a published standard or another implementation — or that prove a parser never reads outside its buffer.

The automatic cycle and its checks (chapter 4): a full collection in a Windows 11 virtual machine, then `check-json.py`, which checks the validity of the outputs and confronts them with one another and with external sources.

Cross-checks on a real collection (chapter 5): what a fresh virtual machine does not contain (old logs, fragmented files, exotic names) is checked on the data of a real machine, processed locally.

1.3 The development rules that go with them

1. **A fix comes with the test that would have caught the defect**, and that test is checked both ways: it passes on the fixed code, and it fails when the defect is reintroduced. A test never seen failing proves nothing.
2. **The build is warning-free**, with `-Wall -Wextra` enabled. The warnings also served as a to-do list: each one marked a field decoded then dropped.

3. **After any change, the new collection is compared with the previous one**, artefact by artefact: an artefact gone from 1,522 entries to 573 lights no warning, only the comparison shows it.
4. **Every input is hostile**, including a file of the examined disk: every read is bounded by the real size of the buffer, never by a length announced in the data.
5. **A field without a value is not emitted**: an empty or null key would read as data found, or as a failed read.

CHAPTER 2

The checks built into WAC

These checks are part of the collection program. They run during every collection, and their result is recorded in `investigation.json`, operation by operation, with its return code and the trace it leaves on the examined system.

2.1 Before any extraction

- Windows version (10 or later) and effective administrator rights (elevated token);
- usable collection location and free space, checked *before* the first extraction: a collection stopped for lack of space would leave an incomplete exhibit store.

2.2 Raw reading of the NTFS volume

Locked files (hives, logs) are read directly from the volume, bypassing the Windows file system. Every structure read is checked before use:

- `FILE` signature of every `$MFT` record and `INDX` signature of every index block, then check of the *fixups* (update sequence array): every 512-byte stride must end with the record's sequence number. A record *torn* by an interrupted write, a mix of two versions, is refused instead of being read as plausible data;
- LZNT1 and WOF decompression bounded by the expected size of each unit: a unit that does not produce the expected number of bytes is an error, not data silently truncated;
- extracted size compared with the size declared by the `$DATA` attribute: their difference, published in the manifest, reveals a truncated extraction a fingerprint alone would not show;
- attributes looked for in the extension records too (`$ATTRIBUTE_LIST`): a directory whose `$INDEX_ROOT` NTFS had moved, the case of thousands of WinSxS components, was reported unreadable (4,569 directories of a Windows 11);
- released index blocks skipped, per the index's `$BITMAP`: they can keep stale entries;
- every attribute read bounded by the attribute's real size, `$STANDARD_INFORMATION` included;
- the source file's dates read in `$STANDARD_INFORMATION` *and* in `$FILE_NAME`: a program forging the former (*timestomping*) leaves the latter untouched, and the gap shows in the manifest;

- volume opened uncached (`FILE_FLAG_NO_BUFFERING`): every read reaches the disk, never an in-memory copy of a sector read earlier; same collection time measured (194 s against 198 s).

2.3 Registry hives

WAC reads the hives with its own reader, linked into the executable, and not with a library of the examined machine.

- base block: `regf` signature, checksum (exclusive OR of the first 127 words), primary and secondary sequence numbers. A hive whose logs have not been replayed is **refused**: read as it is, it would give stale keys without any error;
- replay of the transaction logs (`.LOG1`/`.LOG2`): signature and sequence chaining of every entry; an entry out of the chain is discarded and counted; an undo journal is written before any change, so that the raw copy can be rebuilt. A hive's sequence number is that of the next entry it expects: the entry of the same number is therefore applied, and the header then takes the number after the last entry and the size that entry announces. The result is confronted with Windows: the same hive and its logs, loaded by `reg load`, give the same keys, the same values and the same digest of them all as the hive WAC replayed;
- every offset read in the hive is checked against the file's real size, and a subkey pointing to one of its ancestors is refused: a walk cannot loop on a forged hive.

2.4 Exhibit integrity: the exhibit store

Every extracted exhibit is placed in an *exhibit store* that is never reopened for writing; all the work is done on a second copy.

- three fingerprints per exhibit (MD5, SHA-1, SHA-256), computed while reading the volume;
- the working copy is checked by its SHA-256 against what was read from the volume, not against a re-read of the exhibit store;
- the manifest identifying the exhibits is sealed by a second file carrying its fingerprint: a retouched manifest is detected. Whoever rewrites the manifest can rewrite its seal too: WAC therefore prints the seal at the end of the collection, to be written down in the report, the one copy such a rewrite does not reach;
- a failed extraction is recorded too: an exhibit missing from the manifest would read as never looked for;
- the dates of a file artefact (Prefetch, shortcut, jump list) are the source file's, recorded in the manifest, never those of the working copy, which are the collection's;

- a separate conversion (`--convert`) checks the seal and every exhibit's SHA-256 *before* any read, refuses the store at the first difference, and only accepts exhibit paths that stay inside the store.

A third party can check the exhibit store without WAC, with the scripts `verify-exhibits.ps1` (English messages) and `verifier-consigne.ps1` (French messages), shipped with the user documentation, and their Linux counterparts `verify-exhibits.sh` and `verifier-consigne.sh`, which compute the fingerprints with `sha256sum`, not with WAC; the French ones are the same scripts, started with French messages. All check the seal, and hold it against the value written down at the collection when it is given (`-Seal`, `--seal=`), then the fingerprint of every exhibit. A missing manifest or seal is said absent, not modified; a manifest that is no JSON, or without its list of exhibits, is refused, never taken for an empty collection; an exhibit that cannot be read is said so, apart from the mismatches. The `.ps1` runs from PowerShell 4 (Windows 8.1) on: it reads a manifest of tens of megabytes, which `ConvertFrom-Json` of PowerShell 4 refused, and paths longer than 259 characters, which .NET refuses before version 4.6.2. An authenticated binary read but not copied ("fingerprint only") is counted apart, having no file. The manifest coming from the examined machine, an exhibit path that leaves the collection (`..`, absolute path, symbolic link for the `.sh`) is refused without being read: the first versions of the `.ps1` counted as verified a file of the analysis workstation designated by `exhibits\..\..\..`, provided its fingerprint was in the manifest — a defect caught by a counter-test in the test machine. `verify_exhibits_test.sh` checks the two `.sh` on forged collections (33 checks); the `.ps1` is run in each test machine on its own collection.

2.5 Authenticity of binaries

With the `--binary` option, WAC takes the binaries cited by the artefacts, except those it establishes as signed by Microsoft. That decision rests on a complete verification, done by WAC itself: RSA signature of each security catalog, certificate chain up to a Microsoft root, Authenticode digest of the file found in a verified catalog, or embedded signature. The catalogs read are those of `CatRoot`, of the components (`WinSxS\Catalogs`) and of the servicing packages; the signers accepted, those Microsoft uses for its own code, `.NET` included. The files of a Store application are verified through their package's signature and block map, 64 KiB block by block. A file whose authenticity is not established is collected: doubt leads to collecting, never to discarding. The catalogs, signatures and maps that justified not collecting a file go into the store; every PE carries its build in the manifest (`TimeDateStamp`, `SizeOfImage`), the key of Microsoft's symbol server from which its identical content can be fetched again. With `--binary-all`, everything is collected, and the verdict is recorded for each executable.

2.6 Event logs

Signature of the file, of each chunk and of each record. Every physical chunk of the file is walked, not only those the header declares: a log not properly closed thus yields the records it does not count. An unreadable chunk is reported and skipped; it no longer stops the reading of the file.

Messages: the residual differences with Windows

The cycle compares the message of each kind of event with the one Windows renders itself (`check-json.py`, threshold of 90 % of identical kinds; measured: 887 of 898). The dates inserted are compared without the direction marks (U+200E) Windows puts in them and to the hundred nanoseconds. The same score is reached when the collection (`--collect`) and the conversion are separate. The differences left are explained, kind by kind:

- **a hexadecimal value tied to a map**: Windows does not apply the map as the manifest says. `DehPhase = 0x1` is "PerMachine" by UndockedDeh's manifest, Windows writes "PerUser" (the label of 0); a flags map on `0x2` stays "0x2" with Windows (SMB 30950, 1080). WAC follows the manifest (4 kinds);
- **a satellite older than its binary**: the machine's `fr-FR\audioses.dll.mui` (version 10.0.26100.1) holds these three messages in English, and the faulty text "Signature utilisée :chiffrement" is the one of the `mrxsmb.sys.mui` on the disk; Windows shows a French text absent from those files, whose origin was not found (4 kinds);
- **a null FILETIME**: Windows writes 1601-01-01, WAC inserts nothing (Wininit 25);
- **no message with WAC**: an event whose data Windows could not lay out (`Processing ErrorData`, AppModel-Runtime 219) and Perflib 0 (2 kinds).

2.7 Decoding the artefacts

- every read is bounded by the size of the buffer; an unreasonable number of elements in a list of values (beyond 65,536) is refused rather than walked;
- an impossible date (month 13, hour 25) or one outside the `FILETIME` range gives an absent date, hence not emitted, and never an arbitrary value;
- every local date carries the offset in force *at its date*, according to the daylight saving rules of the examined machine, year by year;
- the SIDs are named from the evidence alone (SAM, well-known SIDs, service SIDs, profiles), never by asking the system or the network: `LookupAccountSidW` queried the domain controller for a SID the machine did not know, a trace of the collection on another machine. The build fails if `WAC.exe` imports a name-resolution function;
- a GUID the reference table does not know is named from the examined machine's hives (machine classes, known folders, each user's classes); unknown everywhere, it is published without a name, never with a placeholder text.

CHAPTER 3

The test programs

Sixteen executables, left out of WAC's build by their `_test.cpp` suffix and built by `build-windows.sh --test`. Each targets one component and confronts it with a judge that shares nothing with it.

3.1 Format harnesses

Program	Judge	Reference result
`sha_test`	FIPS 180-4 vectors, independent implementation	SHA-1 and SHA-256 conform, including lengths 55 to 128 bytes, which exercise the padding
`lznt1_test`	Microsoft's compressor (`RtlCompressBuffer`)	16 compressed units out of 16 conform, 1,048,576 bytes identical
`lzx_test`	Microsoft's compressor (`compact /exe:lzx`)	646 chunks out of 646 matching (three files); 235,707 truncated or damaged inputs without an out-of-bounds read (sanitizers)
`xpress_test`	Microsoft's compressor	137 compressed chunks out of 137 conform, byte for byte
`wevt_test`	files of a real event provider	the whole message production chain checked, substitution marks included
`event_messages_test`	SOFTWARE hive and binaries of the machine	the same chain, step by step, starting from the hive
`evtx_test`	`python-evtx`, independent implementation	record by record; whole chain identical when streamed and in memory (26 MB peak against 1,281 MB)
`offline_registry_test`	Microsoft's `offreg.dll`	18 hives, ten of them from a real machine: 1,273,652 keys and 2,456,139 values identical
`system_conversions_test`	the Windows functions replaced	9,861,570 comparisons, no difference (details below)

Program	Judge	Reference result
`consigne_test`	the exhibit store itself, before and after	exhibit store byte-identical after the working copy was modified; seal conforms; exhibit paths confined to the store
`ntfs_fixup_test`	hand-built records, independent check	sound records restored to the byte, torn records and malformed arrays refused, 20,000 random records against a forbidden page: 44,610 checks, no failure
`mft_test`	`ntfsinfo` (ntfs-3g), an independent NTFS implementation	volume formatted by `mkntfs`: name, parent, size, eight dates and named streams identical for every record; 278 checks, a second volume included, no failure
`usn_journal_test`	the compiler's structures (MinGW's `USN_RECORD`, version 3 from Microsoft)	fields returned as written, numbers equal to the positions in the whole stream, paths through the `$MFT`, damaged records against a forbidden page; 27 checks, no failure. Revealed that the root's sequence was not checked: a parent named by a stale sequence gave a path wrongly asserted
`fat_test` and `vmtest/lib/fat-compare.py`	dissect.fat (Fox-IT); the keys as Windows lists them	FAT12, FAT16 (mkfs.vfat, pyfatfs), FAT32 and exFAT images (formatted by Windows): present entries identical to dissect.fat's, accented long names, deleted entries found; the metadata copy gives the same entries as the volume; a looping FAT chain ends; 17 checks, no failure. Revealed deleted long names cut by a newer entry, now flagged

Program	Judge	Reference result
`ese_test` and `vmtest/lib/ese-compare.py`	dissect.esedb (Fox-IT), an ESE reader written independently; `Get-BitsTransfer`	records built as the format lays them out (fixed, variable, tagged, multi-valued columns), lying lengths refused, BITS blobs; on the VM's `SRUDB.dat` and `qmgr.db` (4 and 16 KiB pages, long values with 64-bit identifiers), 1,064,268 values identical to dissect.esedb's; 19 checks, no failure. Revealed the 8-byte long-value identifiers. On the `Windows.edb` of Windows 8.1 (text compressed on 7 bits), 62,653 identical values; revealed the last character lost of UTF-16 text packed as 7-bit bytes. For 89 values, dissect.esedb decodes one more null character: the padding of the last byte, which the value's header declares
`cim_test`	hash of a name read in the VM's `INDEX.BTR`; `Get-CimInstance`	class, ancestor and instance built as the format lays them out (inherited default, one-byte and UTF-16 strings, property not set), lying lengths refused; on the VM's repository, each subscription identical to what WMI returns; 14 checks, 15 with that repository, no failure
`file_artefacts_test`	each format's published description	PowerShell history, `setupapi.dev.log`, recycle bin `$I` (versions 1 and 2), `hosts`, `Report.wer`, Defender's MPLog: ordinary and hostile cases (truncated `$I`, lying length, impossible date), `Zone.Identifier` streams, DetectionHistory and quarantine entries (RC4 key proven on a header Defender wrote, lying lengths refused); 62 checks, no failure

Program	Judge	Reference result
`shell_items_test`	bytes the VM's Explorer wrote; libfwsi and SBECmd's Lnk library	shellbag items of an archive's content: ZIP (real folder, file with sizes, no date, XP layout) and 7z / tar.gz read through libarchive (real items, UTC date), every truncation against a forbidden page; 683 checks, no failure
`json_test`	WAC's JSON writer	identical round trip, 26 files of a collection read back and rewritten to the byte, malformed texts refused, 200,000 mutations without an out-of-bounds read (sanitizers)
`raw_hive_test`	Windows's `reg load`	the extracted hive is refused before repair and accepted after: the test fails both ways if it must
`authenticode_test`	`Get-Authenticode Signature`	2,233 binaries: 2,126 authenticated by both, 99 collected by both, no file accepted by WAC and rejected by Windows; 462 PowerShell and 11 WSH scripts conform

3.2 Conversions: `system_conversions_test` in detail

WAC depends on five system libraries only (`ADVAPI32`, `KERNEL32`, `msvcrt`, `Secur32`, `WTSAPI32`), needed to query the running system. The conversions it used to ask of other libraries of the examined machine are done by its own code, and this program confronts them with the Windows functions they replace:

- service SIDs, computed offline, compared with `LookupAccountNameW` for the 273 services of the machine;
- GUIDs, SIDs both ways, OLE dates, property names (a table of 1,698 entries generated from the Windows schema);
- FAT dates, on every possible date;
- daylight saving time: offset both ways on the **141 time zones** of Windows, with their rules year by year, at every transition from 1980 to 2035 (hour skipped, hour repeated, change of rule on 1 January);
- reading of the `TimeZoneInformation` key, compared with `GetTimeZoneInformation`;
- time zone rules forged at random: no overflow.

Intended, documented differences: WAC refuses a SID text Windows would truncate, and a non-numeric OLE date for which Windows returns a meaningless time.

3.3 Robustness harnesses

A parser reads at offsets the file itself declares. On a truncated or forged file, a read outside the buffer does not crash on its own: it reads the neighbouring memory and publishes it, in valid JSON. The robustness harnesses make it crash: each input is placed against a forbidden memory page, once at its end, once at its start. A single byte read outside the input stops the test. A watchdog also stops a parser that does not end.

Each file is parsed whole, in its truncations, and in randomly corrupted copies (fixed seed: a failure reproduces identically).

Program	Files of a real machine	Inputs parsed
`lnk_test`	220 shortcuts	546,214
`parsers_test`	48 automatic jump lists	175,585
	23 custom jump lists	79,036
	307 Prefetch files	1,246,491
	4 registry hives	16,207

Result: no read outside the buffer, no endless loop.

CHAPTER 4

The automatic cycle and its checks

4.1 The cycle

`vmtest/run-wac-test.sh` runs, with no interaction at all, a full collection in a Windows 11 virtual machine (real NTFS, real registry):

1. with `--build`, warning-free build of WAC and of the test programs;
2. validation of the raw extraction of a hive: `reg load` must refuse it before repair and accept it after;
3. collection by WAC with SYSTEM rights (`--collect --events --binary`), then conversion of that collection; the exit code is checked, and the trace is searched for any unhandled exception. Before, the bench lays its witnesses: files, registry values, events, browsers, mail, a script whose path passes 260 characters;
4. retrieval of the outputs, of the collection log and of the list of files actually present in the exhibit store;
5. run of `check-json.py`;
6. collection only (`--collect --events --binary`, or `--binary-all`), which must convert nothing; check of the store by `verify-exhibits.ps1`; conversion of a store with one retouched exhibit, which must be refused before any read; conversion, checked by `check-json.py`; second conversion, whose JSON files must be identical to the first's, byte for byte;
7. harmlessness check: two witness files the collection is bound to read keep, before and after it, their three dates and their USN number (`harmlessness.txt`); a control read through Windows changes the access date of one of them, proof that the check can fail. See the *Harmlessness file*;
8. collection driven by a `wac.yml` laid beside WAC, run without any option: it must follow the file (artefacts switched off, no executable) and record it.

4.2 The checks of `check-json.py`

The script first checks the JSON validity of every file and the form of the Windows paths (a double escaping shows after parsing). It then confronts values computed by WAC with independent values: another piece of data of the same collection, or an external source. Each check below was written after a real defect, which it caught.

Check	Defect it caught
Dates	
`X` / `XUtc` pairs, lists included, naming the same instant	double time-zone shift (sessions, shortcuts, installation date); Prefetch run times labelled `+02:00` instead of UTC (1,273 values 2 h off)
offset of every local date / IANA time zone database	every date labelled with the collection day's offset (106 winter dates at `+02:00`); daylight saving dates of the SYSTEM hive read in the wrong layout
no session earlier than the system boot	a two-hour shift on the sessions; later, a boot time 3.5 s late
boot time / Kernel-General 12 event	two independent sources of the same instant: 0.00 s apart
ShimCache date / NTFS date of the file in the manifest	a UTC date treated as local
no event later than the collection	shift or misreading of an event date
FAT dates without a fraction and with even seconds; Amcache text dates without a fraction	every date written with seven digits of fraction, claiming a precision the source does not have (808 dates)
dates of BAM, UserAssist, USBSTOR and Amcache / `taskkill` run by the harness, Prefetch, `Get-PnpDevice Property`, PE headers	these four artefacts store UTC, read as local time: all their dates 2 h off; BAM and UserAssist emptied by a regression of the value reading
Decoding	
output key names in English, without abbreviation	`NbVolumes`, `DateLocale`, `executionTime`: an abbreviation, a French word, a lower-case initial
identifier and serial number of a USB key / `Get-PnpDevice`	no key carried its serial number, vendor or product: they are in the key names, not in values
Prefetch hash / suffix of its file name	65 wrong hashes out of 270 (hexadecimal without padding)
Prefetch `\VOLUME{guid}` paths resolved	20,066 paths out of 20,066 unresolved
Prefetch `$MFT` references / record of the `$MFT` of the same collection (hard links included)	sequence read on four bytes, the entry's high bits lost: 14,940 wrong references out of 14,975

Check	Defect it caught
path and creation date of every exhibit of the system volume in `mft.json` / the exhibit store's manifest (the raw extraction walks the directory indexes down); change journal numbers rising, none after the collection	files with hard links (`System32` and `WinSxS`) appeared under one name only (23,095 exhibits out of 61,242); directories named by their 8.3 alias (`AM3B40~1.945`) when their long name was in an extension record (6,346 exhibits); one exhibit (`sru\SRU00B25.log`, 64 KiB) holding another file's content — a hive's `hbin` blocks —, its record freed by SRUM and reused between the listing of the directory and the read of the file, read without its sequence being checked
witnesses placed by the cycle (PowerShell history, file sent to the recycle bin, `hosts` entry, startup script with its SHA-256) / `powershell_history.json`, `recycle_bin.json`, `hosts.json`, `startup_folders.json`; file marked as downloaded, its stream read back by Windows / `zone_identifiers.json`, read in the `$MFT`; sections of `setupapi.dev.log` counted by Windows / `setupapi.json`; LSA secret names, `PolAdtEv` and `NL$` slots read live by Windows; account rights (`AccountRights`) against `secedit /export` (33 rights) / `security.json`; `REG_EXPAND_SZ` Run value of the DEFAULT hive / `run.json`; persistence witnesses (IFEO, SilentProcessExit, Active Setup, the user's COM hijack) and Windows' own values read live (Userinit, LSA packages, print monitors) / `persistence.json`; second-lot witnesses (trusted document, typed path, search, proxy, firewall rule) and the machine's roots as Windows lists them, each SHA-1 equal to its thumbprint / second-lot JSON	check added with these artefacts
witness program in the machine's `Uninstall` key, and number of programs of each key (machine, 32-bit, user) Windows lists / `installed_programs.json`	showed the users' hives were copied only with the `registry` key: without it, typed paths and searches came out empty

Check	Defect it caught
witness Defender exclusions (path, extension, process) and the list of `Get-MpPreference` / `defender_configuration.json`	check added with the artefact
witness file added to the recent documents by the shell itself (`SHAddToRecentDocs`), first of its list / `explorer_recent_docs.json`	check added with the artefact
witness share mapped by the user, and the `MountPoints2` entries the live registry lists / `mount_points.json`	showed the entries of `CPC\Volume` are volatile, absent from the hive's file
profiles and gateways the live registry lists; last connection in the `NetworkProfile` log (event 10000, UTC) / `known_networks.json`	established these dates are in local time
settings read in the live registry; last shutdown in the System log (Kernel-General 13) / `Operating System.json` (`Settings`, `ShutdownTime`)	check added with these fields
`StartupApproved` and `Run Notification` values read in the live registry / `run.json`	check added with these keys
witness compatibility layer; names of each source read in the live registry; number of pinned items; programs of each `OpenWithList` in `MRUList` order, witness extension included / `program_usage.json`	established `FavoritesResolve` is a sequence of whole shortcuts
USB and SCSI devices and their three dates per `Get-PnpDevice Property` / `devices.json`	check added with the artefact
witness share; shares of `Get-Smb Share` and rights of `Get-SmbShare Access` / `network_shares.json` (decoded security descriptor)	check added with the artefact
adapters of `Get-NetAdapter`: description, hardware address, DHCP address / `network_configuration.json`	check added with the artefact

Check	Defect it caught
logs of `Get-WinEvent -ListLog` and their enabled state / `event_log_configuration.json` (490 logs); sessions of `Get-AutologgerConfig` and their `Start` (39 sessions)	check added with the artefact
Windows 10 `ActivitiesCache.db` (plaso test data, 112 activities) read by Python's `sqlite3`, each column converted independently; the VM's real database, its `-wal` log replayed / `activities_cache.json`	check added with the artefact; a date shifted by one hour is caught (113 failures)
witness document indexed by Windows; Windows' query of its own index (OLE DB, `Search.CollatorDSO`): URL, modification, size of each item / `search_index.json` (515 identical items)	check added with the artefact
pinned public PST (EDRM Enron) placed in the `Outlook Files` of "wac", read by pypff (libpff); Outlook 2013 OST and encrypted PSTs of the java-libpst corpus (`pst_test`) / `emails.json` (the 246 messages pypff reads identical: subject, sender, delivery, 65 attachments by name, size and SHA-256, body; 250 read by WAC)	check added with the artefact
message fetched by Thunderbird itself (local POP3 server, subject in an encoded word, base64 attachment with an RFC 2231 name), mbox read by Python's `mailbox` and `email` modules / `emails.json` (subject and attachment by name and SHA-256 identical)	check added with the artefact; Thunderbird does not fetch without authentication, and in NTLM asks for a password too

Check	Defect it caught
picture seen by ”wac” in Explorer, its `System.ThumbnailCacheId` given by Windows; the two CRC-64 checksums Windows writes for each entry, recomputed / `thumbcache.json` (835 entries of 9 caches, every header checksum and the 777 data checksums identical; the witness’s thumbnail found, exported with its SHA-256)	check added with the artefact
Sysmon 15 installed for the cycle (Windows 11) with a known configuration: a ProcessCreate rule of a witness executable, a FileDelete rule archiving, its log moved to the data volume (`wevtutil sl /lfn`); `Rules` and `ConfigHash` read live by Windows, archives and configuration file hashed by `Get-FileHash`, the moved log read by `Get-WinEvent` / `sysmon.json`, `events.json` (service and driver, SHA-256 of the rules, `ConfigHash` and the file’s SHA-256 identical; every archived file by size and SHA-256; the witness’s ProcessCreate read from the moved log)	check added with the artefact; Sysmon removed after the collections
real RDP session of ”wac” to 127.0.0.2 with a test account, closed as a user would; witness server of known values; cache tiles read by bmc-tools (ANSSI); incoming events listed by `Get-WinEvent` / `rdp_client.json`, `rdp_server.json` (4 tiles with the same pixels, 3 incoming traces, same user and source)	check added with the artefact; the cache is written only when the client closes, and Windows now asks before opening a `.rdp` file

Check	Defect it caught
message rendered by Windows itself (`EvtFormatMessage`, through `Get-WinEvent`) for each kind of event of the VM — provider, identifier, version, qualifiers — on the same record / `EvtEventMessage` of `events.json` (887 kinds of 898 identical)	messages for 28 % of the events only: step of the `WEVT_TEMPLATE` descriptors deduced from the block's size (52 bytes instead of 48); classic sources ignored; UTF-8 tables read byte by byte; `en-US` satellite preferred to `fr-FR`, the language not read in SYSTEM; the data's spaces stripped; values not typed (maps, flags, error codes); data shifted by an array or an empty leaf; manifests moved to `System Resources`; in a separate collection, classic sources and system tables not collected, `CurrentControlSet` never opened
Windows' access audit (read audit rule on the exhibit store of the separate collection, Security events 4663) / reads of the store by `WAC.exe` during the conversion, compared with the instant of its working copy (`conversion.json`): 1,458 reads, all before, none after	some fifteen modules read the exhibits in the store; the `$MFT`, the change journal and the binaries stayed there "read in place"
witness shortcut on the public desktop and shortcut pinned to the task bar (target, arguments, location); shortcuts Windows lists in the Start menus, desktops and Quick Launch / `shortcuts.json` (93 shortcuts with those of the Default profile and of the redirected desktop, no more, no less)	check added with the artefact
Run value planted in the `NTUSER.DAT` of the Default profile / `run.json`, SID `S-0-0-0` and `Profile Default`	check added with the Default profile
NTFS hash of every descriptor of `$Secure:$SDS`, recomputed (2,818 descriptors); owner of 41 files as `Get-Acl` gives it, reached through the `SecurityId` of `mft.json`; Everyone ACE set by `icacls` / `security_descriptors.json`	check added with the artefact

Check	Defect it caught
serial number, sector, cluster and record sizes, clusters of the `$MFT` and of its mirror, identifier, size, delta and first USN of the journal, given by Windows (`DeviceIoControl`) / `ntfs_volumes.json`, read raw in `$Boot` and `$Max` (2 volumes)	check added with the artefact
Defender WPP traces listed by Windows / exhibit manifest, copied raw	check added with the copy
DPAPI master key files listed by Windows, header GUID equal to the name; key each `Preferred` names and its expiry, read by Windows / `dpapi_master_keys.json` (5 keys, 3 folders)	check added with the artefact
values of `UserShellFolders` of “wac” read unexpanded by Windows, its desktop redirected to `OneDrive\Desktop` / `user_shell_folders.json` (20 values); shortcut dropped on the redirected desktop / `shortcuts.json`	check added with the artefact
URLs, visits (URL, instant, transition) and downloads of Edge’s `History`, read again by Python’s SQLite / `web_client.json` (5 URLs, 7 visits)	check added with the artefact
two `TypedURLs` of “wac” and their instants in `TypedURLsTime` / `web_client.json`	check added with the artefact
entries of `WebCacheV01.dat` read by `dissect.esedb` (URL, UTC times; local time of the `MSHist` containers converted by `zoneinfo`) / `web_client.json` (28 entries)	the 2 h shift of `MSHist`’s local time, found before writing the decoder
rows of Edge’s `Login Data` and `Network\Cookies` read again by Python’s SQLite, same sites / `web_client.json` (0 login, 10 cookies)	check added with the artefact

Check	Defect it caught
real browsing of “wac” (Chrome and Firefox installed by `install-browsers.sh`): visit of a witness page, a Google search, a file served as an attachment downloaded by Chrome and by Firefox, an HTA run by `mshta.exe`; Chrome’s databases and Firefox’s `places.sqlite` (WAL replayed) read again by Python’s SQLite / `web_client.json`	check added with the VM’s browsers
session files of Chrome and Edge (`Session_*`, `Tabs_*`) confronted with their `History` read by Python’s SQLite; Firefox’s sessions decompressed by Python’s `lz4` module; `Favicons` and `favicons.sqlite` read by Python’s SQLite / `web_client.json` (108 Chromium session pages: the 57 the history knows carry the instant of a visit, 55 to the microsecond, 2 a navigation within the page 2.5 s later; 54 Firefox pages; 28 page and icon pairs)	check added with the artefact; the format checked on the real files before writing the test
HTA served by the witness server and run by `mshta.exe`: its WebCache entry names its cached copy, collected, same SHA-256 as the content served / `web_client.json`, manifest	check added with the cache’s collection
records of four real `index.dat` files (plaso’s test data) read by `pymsiecf` / the `index.dat` decoder (1,110 records identical)	a recovered record partly overwritten gave another’s data
download by the real Internet Explorer 11 (COM object, “Save”, normal close) / `web_client.json` (`iedownload`: URL, saved path, size, server address)	the size and address read at wrong offsets, which the unit test, built on the same ones, could not see
DestList entries / jump list shortcuts	the first 15 entries of every list lost (573 shortcuts published out of 1,522)
machine name of the events / `OperatingSystem.json`	check of the BinXML decoding, on two unrelated sources

Check	Defect it caught
record identifiers unique per log	a stale chunk read again
unresolved `%%nnnn` references in messages	a provider's parameter file not loaded (3,780 messages)
RID found at the end of the rebuilt SID	reading of the SAM database
NT hash of a witness account with a known password, the previous one in the history / `users.json`	MD4 of the password in UTF-16LE, recomputed apart (the boot key / AES or RC4 / DES chain verified end to end, on Windows 11 and on Windows 8.1; on 8.1, every hash was wrong: a constant of the RC4 scheme had one "Q" too many)
direct members of each local group, witness account in Administrators / `local_groups.json`	`Get-LocalGroupMember` over the 21 groups
security questions (`Security Questions`) of a witness account / `users.json`	`ResetData` written then read back, questions and answers
service states and types, presence of drivers	enumeration filters mistaken for states
no process carries `WAC.exe` among its modules	the tool's modules attributed to the Idle process
plausible `$MFT` references	decoding of the shortcuts' extension blocks
mounts / Windows' reading of the key (`reg query`) and partitions (`Get-Partition`)	6 mounts out of 7 published as "\" (device paths decoded as ANSI)
SID names / `Get-LocalUser`, `Get-LocalGroup`	names asked of the running system, even of the domain controller, and translated into the workstation's language ("Système")
GUIDs unknown to the table / three GUIDs registered by the harness, one per source	"Unmapped GUID" published as a name, among them 18 COM handlers of the VM's tasks, named in its hive
Exhibit store and replay	
seal / real fingerprint of the manifest	a retouched manifest
three fingerprints per exhibit, consistent counts	an unidentified exhibit
files present in the exhibit store / manifest	209 exhibits added after sealing, which every other check let through
no hive both replayed and repaired	a replay that does not align the sequences
an undo journal for every replay	a raw copy impossible to rebuild
Raw reading, separate collection and conversion	

Check	Defect it caught
SHA-256 of files read raw / `Get-FileHash`	safeguard of the reading by batches of contiguous clusters
Prefetch dates / dates given by Windows	0 out of 335 matching: dates of the working copy, the minute of the collection
executables of System32 and SysWOW64 / manifest of a `--collect --binary`	6,446 missing out of 8,540 (`$INDEX_ROOT` in an extension record, names of hard links not declared)
catalogs cited in a verdict / exhibits of the store	170 out of 170 missing (label “catalog” looked for as “catalogue”)
WAC’s Microsoft verdicts / `Get-AuthenticodeSignature`	safeguard: a wrong verdict would leave a binary out of the store
copy of a Store package: untouched, modified, added file	verification through the signed block map
dates of a script forged back to 2001 / `$STANDARD_INFORMATION` and `$FILE_NAME`	safeguard: a forgery must stay visible
retrieval keys / PE headers read by PowerShell	safeguard of the retrievable content
`--binary-all`: copy and verdict of every executable	package check too strict under `--binary-all`
retouched exhibit / refusal of the conversion	safeguard of the check before reading
two conversions of one collection	dates read on the working copy, rebuilt at each conversion

4.3 Reference result

Last cycle, 9 October 2026, on version 2.0.0 (Windows 11, Paris time zone): 472 checks passed, no failure.

- **collection then conversion** (`--collect --events --binary`, collection 7 min, conversion 1 min 47 s): 72 valid JSON files, USBSTOR included (the harness attaches two virtual USB keys on every cycle, so that a recreated test machine knows them too); 226 checks passed, among them 59,107 local dates confronted with the IANA database and 211,972 events decoded from 405 logs, 98.5 % of the event kinds with the text Windows renders; 6,014 files in the exhibit store, exactly those of the manifest, seal conforms; a script whose path is 261 characters long recorded with the fingerprint Windows gives;
- **collection only** (`--collect --events --binary`, 7 min 30 s): nothing converted; 49,226 executables read, 28,180 authenticated as Microsoft and 328 third-party cleared by

their chain, not copied; 5,430 contents already recorded; 2,597 collected (771 MB); 11,515 exhibits verified by `verify-exhibits.ps1`; a retouched exhibit refused before any read; harmlessness check conforms;

- **conversion** (`--convert`, 1 min 43 s): 243 checks passed, among them the 8,537 executables of System32 and SysWOW64 present in the manifest, 625 retrieval keys equal to the PE headers, a timestomped script visible (`$STANDARD_INFORMATION` of 2001, `$FILE_NAME` of the day), the Store package verification; a second conversion gives identical JSON files, byte for byte;
- **collection driven by `wac.yml`**: the file is followed and recorded.

CHAPTER 5

Cross-checks on a real collection

A fresh virtual machine produces neither fragmentation, nor old jump lists, nor exotic file names, nor damaged logs. Several defects only showed on a real machine, confronted with an independent source:

Cross-check	What it showed
jump lists / `olefile`	the streams are named “1” to “f”, not “01” to “0f”
shortcut target / DestList entry	a wrong code page and an ignored Unicode path
logs / `python-evtx`	a chunk with a wrong signature stopped the reading of the whole file (14 records read out of 270)
EVTX chunks / their CRC32	one-off check: the 1,505 chunks of the 404 logs of the test machine are intact
hives / `offreg.dll`	1,273,652 identical keys
binaries / `Get-AuthenticodeSignature`	no undue authentication
old output / new one	the same files decoded by the current code, compared after normalisation

> **Keep in mind**
> A real collection contains personal data. It is processed locally; a temporary copy in the virtual machine or in a working folder is deleted after use, and nothing of it enters the source repository.

CHAPTER 6

What these checks guarantee

6.1 Guarantees

- **Exhibit integrity**: every exhibit is identified by three fingerprints, the working copy is checked against the reading of the volume, the manifest is sealed, and the exhibit store is never modified (`consigne_test`, checks of the cycle).
- **No writing to the examined system**, apart from the unavoidable traces of a collection on a running system, all recorded in `investigation.json` with the expected trace.
- **Accuracy of the decoders**: where a reference exists — hive reader, decompressions, fingerprints, conversions, time zones, authenticity of binaries — results identical to it on the volumes stated.
- **Robustness**: no shortcut, jump list, Prefetch or hive parser reads outside its buffer or loops, on more than two million truncated or corrupted inputs.
- **Consistency of the outputs**: every possible cross-check between two artefacts, or between an artefact and an external source, is run in every cycle.
- **Independence from the examined machine**: WAC borrows no library from it to interpret the evidence.

6.2 Limits

> **Limit**
>
> These checks do not prove the absence of defects. They prove that the defects they target are absent, on the data they have seen.

- **Completeness**: an empty artefact cannot be told apart from a machine without traces. Only the comparison between two collections, or with a machine whose content is known, measures it.
- **Data the test machine does not hold**: MTP devices, files spread over several `$MFT` records (`$ATTRIBUTE_LIST`), very large volumes. They are to be validated on real systems.
- **Artefacts without a reference**: when no independent tool reads a format, its accuracy rests on the internal cross-checks and on the public documentation of the format.
- **Time zones**: the confrontation with the IANA database covers the zones whose rules agree with Windows's over the period checked (Western Europe, United States, UTC); the others are checked against Windows only.

- **Building on Windows**: `build-windows.cmd` runs `build-windows.sh` in Git for Windows' bash, with MinGW-Builds' gcc 13.2.0 ("win32" thread model, msvcrt). The full cycle was passed once with the executable built this way, with no difference from the Linux build; the routine cycles are made on the Linux build, the one delivered. A gcc with the "posix" thread model (MSYS2) is refused: it adds 38 imports to `WAC.exe`.